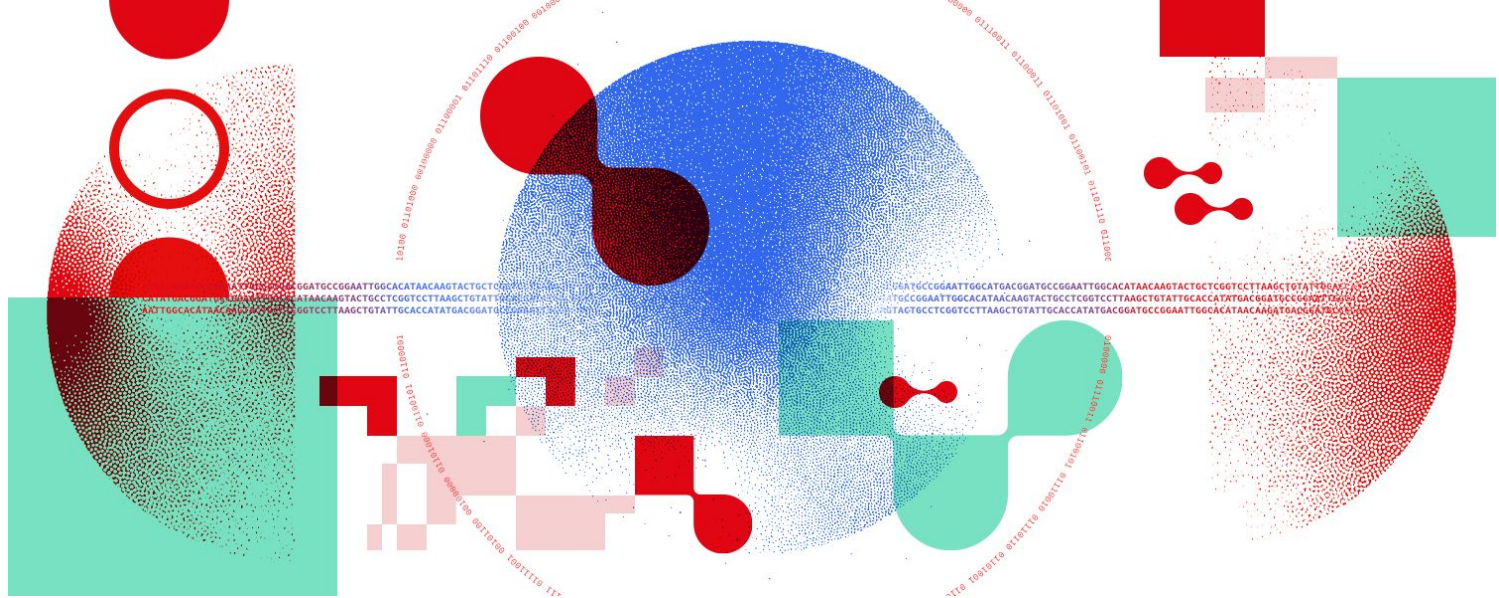




Snakemake for reproducible research

Making Snakemake even more reproducible

What could we improve? (again)

- Being reproducible
- Using scripts from other languages
- Using unknown number of **inputs/outputs**

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- Being reproducible —————→ conda/mamba, Docker/Apptainer
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Being reproducible with Snakemake and Conda

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Being reproducible with Snakemake and Conda

- What is conda?
 - **Conda/mamba**: open-source, cross-platform, language-agnostic package manager and environment management system
 - **Channels**: field-specific repositories of software
 - [Conda-forge](#): general computation
 - [Bioconda](#): bioinformatics

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 - **Conda/mamba**: open-source, cross-platform, language-agnostic package manager and environment management system
 - **Channels**: field-specific repositories of software
 - [Conda-forge](#): general computation
 - [Bioconda](#): bioinformatics
 - Environments defined in YAML files

```
name: python_env
channels:
  - conda-forge
  - bioconda
dependencies:
  - python >= 3.12
  - pandas == 2.2.3
```

py.yaml

Being reproducible with Snakemake and Conda

- Using conda in Snakemake:
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```
rule rename_file:
    input:
        'data/test.txt'
    output:
        'results/renamed_file.txt'
    conda:
        '../envs/py.yaml'
    shell:
        'mv {input} {output}'
```

Being reproducible with Snakemake and Conda

- Using conda in Snakemake:
 - Snakemake provides integrated package management via Conda to define isolated software environments per rule
 - **conda directive:**
 - Value: path to the environment file relative to the rule's snakefile
 - Execution parameter:
 - v7 and before: `--use-conda`
 - v8+: `--software-deployment-method` or `--sdm` (shorthand version)

```
rule rename_file:
    input:
        'data/test.txt'
    output:
        'results/rename_file.txt'
    conda:
        '../envs/py.yaml'
    shell:
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```

```
snakemake --cores 1 --use-conda results/rename_file.txt
```

```
snakemake --cores 1 --sdm conda
results/rename_file.txt
```

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 - Value: URL/path to the image location
 - Handles Docker and Apptainer images
 - **Global OR rule-specific**

```
container:
'docker://athiebaut/deseq2:v3'

rule rename_file:
    input:
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        'mv {input} {output}'
```

```
snakemake -c 1 --use-singularity
results/renamed_file.txt
```

```
snakemake -c 1 --sdm apptainer
results/renamed_file.txt
```

Being reproducible with Snakemake and Docker

- Using Docker in Snakemake:
 - Snakemake provides a container integration: it can automatically spawn a container created from a given image
 - **container directive:**
 - Value: URL/path to the image location
 - Handles Docker and Apptainer images
 - **Global OR rule-specific**
 - Execution parameter
 - v7 and before: `--use-singularity`
 - V8+: `--sdm apptainer`
 - Can be combined with conda `--sdm conda apptainer`
 - Pull the image
 - Create the conda env from **within the container**
 - Containerisation of Conda-based workflows

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container:
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rule rename_file:
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snakemake -c 1 --use-singularity
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```

```
snakemake -c 1 --sdm apptainer
results/renamed_file.txt
```

```
snakemake -c 1 --sdm conda apptainer
results/renamed_file.txt
```

```
snakemake -c 1 --containerize > Dockerfile
```

What is the best setting for Snakemake environments?

- Use package and container managers!
- Same as Snakefile and config files: split things reasonably
 - 1 .smk file $\approx$ 1 'thematic' module $\approx$ 1 environment
- Always check for version conflicts

What could we improve? (again)

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Executing external code in Snakemake

- There are 2 ways to execute external code in Snakemake: `run` and `script`

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```
rule get_header:
    input:
        'data/file.txt'
    output:
        'results/file_header.txt'
    params:
        lines = 5
    run:
        input_file = open(input[0])
        output_file = open(output[0], 'w')
        for i in range(params.lines):
            output_file.write(input_file.readline())
```

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    input:
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- Replaces **shell/run**
- Access to directive values and variables, like in **shell**
- Value = path to the script relative to the rule's snakefile
- Advantages:
 - Great for long code
 - Can use **conda/container** directive!!!

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    params:
        lines = 5
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```

```
# Retrieve information from Snakemake
input_file = open(snakemake.input[0])
output_file = open(snakemake.output[0], 'w')
n_lines = snakemake.params.lines

# Process file
for i in range(n_lines):
    output_file.write(input_file.readline())
```

first_step.py

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    script:
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```

```
library(readr)

# Retrieve information from Snakemake
input_path <- snakemake@input[[1]]
output_path <- snakemake@output[[1]]
n_lines <- snakemake@params$lines[1]

# Process file
data <- read_delim(input_path, '\t',
```

first_step.R

What could we improve? (again)

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Working with an unknown number of inputs/outputs

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 - **Input** files depend on wildcards
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- How to use an input function?
 - Define the function above the rule
 - Use the syntax `input: <function_name>`
 - No parentheses, no argument

```
def seq_input(wildcards):  
    type = wildcards.type  
    if type == 'SE':  
        return 'data/file1.fq'  
    else:  
        return ['data/file1.fq', 'data/file2.fq']  
  
rule merge_files:  
    input:  
        seq_input  
    output:  
        'results/{type}.txt'  
    shell:  
        'cat {input} > {output}'
```

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- Input functions = Python functions
 - Single argument: **wildcards**
 - Return a file or list of files
 - Can also return a dictionary with **input** names as keys
 - Use `input: unpack(<function_name>)` to obtain named inputs

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def seq_input(wildcards):  
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- Input functions = Python functions
 - Single argument: **wildcards**
 - Return a file or list of files
 - Can also return a dictionary with **input** names as keys
 - Use `input: unpack(<function_name>)` to obtain named inputs
- Functions are evaluated before workflow execution $\Rightarrow$ can't list **output** files
 - **No output functions!**

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    type = wildcards.type  
    if type == 'SE':  
        return 'data/file1.fq'  
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rule merge_files:  
    input:  
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    output:  
        'results/({type}).txt'  
    shell:  
        'cat {input} > {output}'
```

```
snakemake --cores 1 results/not_SE.txt
```



```
{type} = "NotSE"
```



```
input:  
    ['data/file1.fq', 'data/file2.fq']
```

Working after an unknown number of inputs/outputs

- *aka* 'Data-dependent conditional execution' *aka* **checkpoint** (instead of **rule**)
- When:
 - An unknown number of files is generated by a rule
 - **Output** files are unknown before execution
- Conditional reevaluation of the DAG of jobs based on the outputs content
 - Since DAG is re-evaluated midway □ you can't see the whole workflow at the start
- **Very complicated!**

Exercises

- Through the day:
 - Develop a simple RNAseq analysis workflow, from reads (fastq files) to Differentially Expressed Genes (DEG)
- For this session:
 - Use an input function
 - Use R and Python scripts
 - Run a rule inside a conda environment
 - Run a rule inside a Docker/Singularity container

